Stage 2 Implementation Report

AI Code Review Agent

EECS3311 — Course Project, Stage 2 (AI Human Collaborative Implementation)

Project	AI Code Review Agent
Repository	`https://github.com/dev91299/EECS3311-course-project`
Team member(s)	`Devraj Patel (221059118)`
Submission date	10 November 2026
Implementation period	20 October – 10 November 2026
Language / toolchain	Java 17+ (compiled and tested on OpenJDK 21), Maven, zero runtime dependencies
Entry point	`com.eecs3311.codereview.app.Main`

1. Project overview and implementation status

1.1 What the application is

The AI Code Review Agent is a desktop tool that reviews source code. A developer points it at a repository; the agent indexes it, and can then explain code, detect defects and grade them by severity, check a project coding standard, propose refactorings, write unit tests, answer questions about the codebase in natural language, compare review results across two versions, and export a report.

Beyond those request-response features, the agent can be given a goal in plain language and left to pursue it on its own: it writes a plan, chooses which of its tools to invoke, reads the code it decides is relevant, records findings, and writes a synthesis when it judges the goal met.

1.2 Users

- **Software developers** reviewing their own work before opening a pull request.
- **QA engineers** looking for defects and missing test coverage.
- **Project leads** who need an exportable record of what was found and what the team decided about each finding.

1.3 AI and LLM integration

The system speaks the OpenAI `/chat/completions` protocol, so a single transport class covers four backends, selected by one line of configuration:

`LLM_PROVIDER`	Endpoint	Default model	Cost
`groq` *(default)*	`https://api.groq.com/openai/v1`	`llama-3.3-70b-versatile`	Free tier, no credit card

LLM_PROVIDER	Endpoint	Default model	Cost
openrouter	https://openrouter.ai/api/v1	google/gemma-4-31b-it:free	Free tier
ollama	http://localhost:11434/v1	llama3.1	Free, runs locally
openai	https://api.openai.com/v1	gpt-4o-mini	Paid

The model is reached through `LLMClient`, the single point of access, which wraps the transport in four decorators: retrying, caching, rate limiting and usage logging.

Where AI is actually used. Each call is labelled with a purpose, and the logging decorator accumulates tokens per purpose, so the system can report exactly what the model was used for: `security-analysis`, `style-analysis`, `performance-analysis`, `maintainability-analysis`, `code-summary`, `test-generation`, `refactor-suggestion`, `chat-assistant`, `delta-summary`, `agent-planning`, `agent-decision` and `agent-synthesis`.

1.4 Why this qualifies as meaningful agent behaviour

The Stage 2 instructions state that forwarding a prompt to an LLM API and displaying the response is insufficient. This implementation goes beyond that in six specific ways.

Required capability	Where it is implemented	What actually happens
Planning	`AgentPlanner.createPlan`	The goal plus the real repository inventory are turned into an ordered plan naming actual files. The plan is grounded: the planner is shown what exists so it cannot propose reading files that are not there.
Tool use	`ToolRegistry`, five `AgentTool` implementations	The agent selects a tool by name and passes it a string argument. It never executes model output as code. Tool descriptions are generated from the registry, so the catalogue and the prompt cannot drift apart.
Retrieval	`CodeRetriever`	A TF-IDF-weighted index over the repository. Questions and analyses are answered against the few regions that actually matter, which is what lets the assistant work on a codebase far larger than any context window.
Memory	`AgentMemory`	Short-term: the run's observations, re-read before every decision. Long-term: the team's accept/reject decisions, persisted to `.agent-memory/memory.json` and recalled by relevance into later prompts.

Required capability	Where it is implemented	What actually happens
Decision making	`AgentPlanner.decideNextAction`	One action per turn, chosen from everything observed so far. A repeated action is blocked and the block is fed back as an observation the agent must react to. A tool error becomes an observation rather than terminating the run.
Multi-step execution	`AgentPlanner.run`	Plan → decide → act → observe → repeat → conclude, bounded by a configurable step budget so a run always terminates. Every step is published to both views as it happens.

A concrete example from a real run against `samples/demo-project`:

```
[step 1] PLAN         1. Call list_files to take an inventory of what the repository contains.
                      2. Call search_code to locate the code that handles credentials.
                      3. Call analyze_file with the security strategy on each file found.
[step 2] THOUGHT      Inventory the repository before choosing targets.
[step 2] ACTION       list_files
[step 2] OBSERVATION  Repository contains 3 source files ...
[step 3] THOUGHT      Locate the code most likely to handle credentials.
[step 3] ACTION       search_code: password authentication query
[step 3] OBSERVATION  Found 1 relevant region: LoginService.java (from line 26) ...
[step 5] ACTION       analyze_file: src/com/acme/shop/LoginService.java|security
[step 5] OBSERVATION  recorded 5 finding(s): [Critical] line 15: Credential hard-coded ...
[step 8] CONCLUSION   Audit complete
```

The agent is also **adaptive**. Rejecting a finding with a reason writes that decision to long-term memory; the next analysis of related code carries an instruction derived from it:

> `- The team REJECTED the finding "Division without a zero guard", saying: sessionCount is validated by the caller. Do not report this again unless the circumstances clearly differ.`

1.5 Implementation status

Fully implemented and working: all thirteen features in section 2, both interfaces, ten design patterns, the LLM integration with four interchangeable backends, and the autonomous agent loop. All ten features proposed in Stage 1 are implemented; none were dropped.

Verification: 110 assertions pass in the dependency-free harness (`./build.sh --check`), plus a JUnit 5 suite (`mvn test`). Both the GUI and the CLI have been exercised end-to-end, and the real HTTP path has been verified against a mock OpenAI-compatible endpoint covering success, malformed output, a 401 and a retried 503.

Honest limitations

These are real and are not worked around:

1. **Output quality depends on the model.** On a free-tier 70B model, findings are good on obvious defect classes (injection, hard-coded secrets, swallowed exceptions, missing guards) and weaker on subtle cross-file logic errors.

The system constrains the model — a fixed JSON schema, a severity ceiling per strategy, line-number validation — but it cannot make a small model reason better.

2. **Offline mode is deliberately limited.** With no API key the application still runs, using a built-in static analyser that derives findings from the code by regular expression. It is a fallback so the interface and the test suite work without a key; it does not reason, and both interfaces display an OFFLINE MODE banner whenever it is active. **Every AI-based feature in the demonstration videos is shown running against a real model.**
3. **Analysis is per-file.** The retriever supplies related snippets as context, but a defect that only exists in the interaction between three classes may be missed.
4. **F08 compares snapshots, not arbitrary commits automatically.** The user analyses one version, snapshots it, checks out the other, analyses it and snapshots that. Git provides the changed-file list; the tool does not check out commits on the user's behalf, which is deliberate — silently changing a developer's working tree would be unsafe.
5. **Refactorings are suggestions, never applied automatically.** The tool shows a side-by-side comparison and offers to copy the result. It does not write to source files.
6. **No authentication or multi-user support.** It is a single-user desktop tool.

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2. Feature to implementation mapping

Every Stage 1 feature (F01–F10) is implemented. Three features (F11–F13) were added during implementation; they are justified in section 4.

F01 — Repository Import and File Browsing

Mapping field	Entry
Feature ID and name	F01 — Repository Import & File Browsing
Stage 1 feature	F01
Status	Implemented (extended: Git cloning added)
Implemented classes/files	`repo/RepoManager.java` — all file-system and Git access, and the sandbox boundary; `model/FileNode.java` — the Composite tree; `app/AgentController.java` — orchestrates import; `ui/CodeReviewGUI.java` — JTree; `ui/CLIView.java` — `open`, `tree`, `files`, `cat`
Key methods	`RepoManager.loadFiles(String)`, `RepoManager.cloneAndLoad(String)`, `FileNode.getTotalFiles()`, `FileNode.flatten()`, `FileNode.find(String)`, `AgentController.handleImport(String, boolean)`
Design patterns	Composite
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A`

Type: Deterministic.

How the classes collaborate. The user chooses a folder in the GUI, or types `open <path>` in the CLI. Either way the view calls `AgentController.handleImport`. The controller delegates to `RepoManager.loadFiles`, which validates the path — reporting "does not exist", "is a file, not a directory" or "permission denied" distinctly, which is the alternative flow use case UC01 specifies — then walks the directory, skipping build output and dependency folders, and builds a `FileNode` tree. The controller then clears the previous session, calls `ReviewAgent.onRepositoryImported` (which loads the project's `.rules` file, builds the retrieval index and attaches long-term memory to this repository), and publishes `REPOSITORY_LOADED`. Both views receive that event and render the tree.

F02 — Code Summarisation

Mapping field	Entry
Feature ID and name	F02 — Code Summarisation
Stage 1 feature	F02
Status	Implemented
Implemented classes/files	`agent/ReviewAgent.java` — Facade, builds the prompt; `llm/PromptBuilder.java` — prompt assembly; `llm/LLMClient.java` — the single model entry point; `ui/*` — invoke and display
Key methods	`ReviewAgent.summarise(String, String)`, `PromptBuilder.build()`, `LLMClient.sendPrompt(String, String, String)`
Design patterns	Facade, Singleton, Builder
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=75`

Type: AI-based.

How the classes collaborate. The user selects a file, or highlights part of one, and presses **Summarise** (`summarize` in the CLI). `AgentController.summarise` passes the selection to `ReviewAgent.summarise`, which uses `PromptBuilder` to assemble a prompt asking for four specific sections — purpose, how it works, dependencies and risks — rather than an open-ended "summarise this", because an unstructured request produces a paraphrase of the code instead of an explanation of it. `LLMClient` sends it; the reply is displayed in the AI output panel. If the selection exceeds the context budget the builder truncates it with an explicit marker and the user is told to select less, which is the error case F02 specifies.

F03 — Issue Detection and Severity Categorisation

Mapping field	Entry
Feature ID and name	F03 — Issue Detection & Severity Categorization
Stage 1 feature	F03
Status	Implemented (extended: four strategies instead of one, severity is now a type-safe enum)

Mapping field	Entry
Implemented classes/files	`agent/strategy/ReviewStrategy.java`, `AbstractReviewStrategy.java`, `SecurityStrategy.java`, `StrategyRegistry.java`; `model/ReviewItem.java`, `model/Severity.java`, `model/ReviewReport.java`
Key methods	`AbstractReviewStrategy.executeStrategy(AnalysisContext)` *(final template method)*, `SecurityStrategy.decoratePrompt(...)`, `AbstractReviewStrategy.parseFindings(...)`, `AbstractReviewStrategy.validate(...)`, `ReviewReport.addItems(List)`
Design patterns	Strategy, Template Method, Facade, Singleton, Observer
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=100`

Type: AI-based.

How the classes collaborate. Pressing **Analyse** calls `AgentController.requestReview(path, code, strategyName)`. The controller delegates to `ReviewAgent.analyzeCode`, which first assembles an `AnalysisContext` containing the code, the project's conventions, the recalled team decisions from `AgentMemory`, and related snippets from `CodeRetriever` — all gathered *before* any model call. `StrategyRegistry` resolves the named strategy and `executeStrategy` runs the fixed seven-step template: reject empty input, run the deterministic pre-pass, assemble the prompt, call the model once, parse the reply, validate it, post-process. `parseFindings` tolerates the three ways models return JSON badly (fenced, wrapped in prose, or buried in a wrapper object). `validate` corrects an out-of-range line number to file-wide rather than discarding an otherwise valid finding, and clamps severity to the strategy's ceiling. Findings are added to `ReviewReport`, which notifies both views.

F04 — Refactoring Suggestions

Mapping field	Entry
Feature ID and name	F04 — Refactoring Suggestions
Stage 1 feature	F04
Status	Implemented
Implemented classes/files	`agent/ReviewAgent.java`; `util/Diff.java` — LCS diff; `app/AgentController.java` — `RefactorResult`; `ui/CodeReviewGUI.java` — side-by-side dialog; `ui/CLIView.java` — text columns
Key methods	`ReviewAgent.suggestRefactor(String, String, ReviewItem)`, `AgentController.suggestRefactor(...)`, `Diff.compute(String, String)`, `Diff.renderSideBySide(...)`, `Diff.countChanges(...)`
Design patterns	Facade, Singleton, Builder
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=135`

Type: AI-based.

How the classes collaborate. The user presses **Refactor**, or selects a finding and presses **Fix this**. When a finding is given, its severity, title, description and line are written into the prompt so the rewrite targets that specific defect. The model is required to return the complete rewritten version in a fenced block followed by `WHY:` bullets. The controller extracts the code block; if the reply contains no usable block it requests one more time — the "malformed response, request regeneration" error case from Stage 1 — and fails with a clear message if the second attempt also fails. `Diff.compute` runs a longest-common-subsequence diff so an inserted line does not mark everything after it as changed, and the GUI shows original and suggested side by side with the change counts. **Nothing is written to the user's source files.**

F05 — Test Generation Suggestion

Mapping field	Entry
Feature ID and name	F05 — Test Generation Suggestion
Stage 1 feature	F05
Status	Implemented
Implemented classes/files	`agent/ReviewAgent.java` ; `llm/PromptBuilder.java` ; `ui/*`
Key methods	`ReviewAgent.generateTests(String, String)` , `ReviewAgent.detectTestFramework(String)`
Design patterns	Facade, Singleton, Builder
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=168`

Type: AI-based.

How the classes collaborate. `detectTestFramework` picks a framework from the file's language — JUnit 5 for Java, pytest for Python, Jest for JavaScript and TypeScript, NUnit for C# — or honours an explicit `TEST_FRAMEWORK` setting. This is the fallback use case UC03 describes for an unrecognised framework: the system defaults to the standard library framework for the language rather than failing. The prompt requires coverage in a stated order (typical case, each boundary, each failure path, null and empty input), one behaviour per test, real assertions rather than placeholders, and a closing list of anything the model could not test and why. The result is shown for the developer to copy; the tool does not write test files.

F06 — Coding-Standard Checking

Mapping field	Entry
Feature ID and name	F06 — Coding-Standard Checking
Stage 1 feature	F06
Status	Implemented

Mapping field	Entry
Implemented classes/files	`repo/StandardsChecker.java` — the deterministic rule engine; `agent/strategy/StyleStrategy.java` — the hybrid strategy; `util/SourceText.java` — comment and string blanking; `samples/demo-project/.rules` — example rule file
Key methods	`StandardsChecker.loadRules(String)`, `StandardsChecker.check(String, String)`, `StandardsChecker.describeRules()`, `StyleStrategy.runDeterministicPass(AnalysisContext)`
Design patterns	Strategy, Template Method
Video link/timestamp	`https://youtu.be/AqS-KN2L4fM?t=25`

Type: Hybrid — this is the clearest example of the split.

How the classes collaborate. On import, `RepoManager.readRulesFile` looks for `.rules`, `.codereview-rules` or `CONVENTIONS.md`; if none is present the built-in defaults apply, which is the "missing rule file" fallback F06 specifies. When the style strategy runs, `StyleStrategy.runDeterministicPass` — the Template Method's deterministic hook — calls `StandardsChecker.check`, which decides every rule that can be decided exactly: line length, tabs, wildcard imports, unbraced conditionals, constant and type and method casing, parameter count and method length. Only then is the model prompted, and `describeRules()` injects the active rules together with an explicit instruction **not** to repeat them, so the two halves do not both report the same violation. The model is asked only about judgement calls: whether an identifier describes what it holds, whether a comment adds information, whether a class has taken on a second responsibility.

Pattern matching runs over a comment-and-string-blanked copy of the file produced by `SourceText.stripCommentsAndStrings`, with character positions preserved so line numbers stay correct. Section 6.3 records the defect that made this necessary.

F07 — Review Acceptance/Rejection and Human Feedback Capture

Mapping field	Entry
Feature ID and name	F07 — Review Acceptance/Rejection & Human Feedback Capture
Stage 1 feature	F07
Status	Implemented (extended: undo, re-grading and bulk triage added)
Implemented classes/files	`command/ReviewCommand.java`, `AcceptCommand.java`, `RejectCommand.java`, `ChangeSeverityCommand.java`, `BulkTriageCommand.java`, `CommandHistory.java`; `model/ReviewItem.java` (Memento); `agent/AgentMemory.java`

Mapping field	Entry
Key methods	`AcceptCommand.execute()/undo()`, `RejectCommand.execute()/undo()`, `ReviewItem.createMemento()`, `ReviewItem.restore(Memento)`, `CommandHistory.execute(ReviewCommand)`, `AgentMemory.recordDecision(ReviewItem)`
Design patterns	Command, Memento, Observer
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=180`

Type: Deterministic.

How the classes collaborate. The GUI's Reject button and the CLI's `reject <ID> <reason>` both reach `AgentController.rejectFinding`, which constructs a `RejectCommand` and hands it to `CommandHistory.execute`. The command captures a memento of the finding's current triage state, applies the change, and notifies `ReviewReport`, which pushes `ITEM_CHANGED` to every view. The controller then calls `ReviewAgent.rememberDecision`, which writes the decision and its reason into `AgentMemory` and persists it to `.agent-memory/memory.json`. That file is what makes the agent adapt: the next analysis of related code recalls the decision by relevance and includes it in the prompt as an instruction.

Sequence diagram **SD02** shows this flow including the undo path.

F08 — Comparison of Reviews Across Versions

Mapping field	Entry
Feature ID and name	F08 — Comparison of Reviews Across Versions
Stage 1 feature	F08
Status	Implemented
Implemented classes/files	`model/ReviewSnapshot.java`, `model/DeltaReport.java`; `repo/RepoManager.java` — Git access; `agent/ReviewAgent.java` — the prose verdict
Key methods	`ReviewReport.snapshot(String, String)`, `DeltaReport` constructor, `ReviewItem.getFingerprint()`, `RepoManager.getDiff(String, String)`, `RepoManager.getChangedFiles(...)`, `AgentController.compareSnapshots(String, String)`, `ReviewAgent.summariseDelta(DeltaReport)`
Design patterns	— (Facade and Singleton are used for the AI half)
Video link/timestamp	`https://youtu.be/AqS-KN2L4fM?t=105`

Type: Hybrid.

How the classes collaborate. The user analyses one version and saves a snapshot, then analyses the other and saves a second. `DeltaReport` splits the two finding sets into resolved, newly introduced and still-present, using

`ReviewItem.getFingerprint()` — file, category and headline, deliberately **excluding the line number**, so inserting a line above a defect does not make the old issue look resolved and a new one invented. That arithmetic is fully deterministic and is the authoritative result. If both snapshots carry commit hashes and the folder is a Git working tree, `RepoManager.getDiff` adds the changed-file list. Only then is the model asked, through `summariseDelta`, to interpret the direction of travel. If that call fails the deterministic comparison is still shown, with a note that the AI assessment was unavailable. An unknown commit produces a "Git Reference Error", the error case Stage 1 specified.

F09 — Review-History Tracking and Report Generation

Mapping field	Entry
Feature ID and name	F09 — Review-History Tracking & Report Generation
Stage 1 feature	F09
Status	Implemented (extended: four formats instead of two)
Implemented classes/files	`export/ReportExporter.java`, `AbstractReportExporter.java`, `PdfExporter.java`, `MarkdownExporter.java`, `HtmlExporter.java`, `JsonExporter.java`, `ExporterFactory.java`, `SimplePdfWriter.java`
Key methods	`AbstractReportExporter.export(ReviewReport, Path)` *(final template method)*, `ReviewReport.aggregateItems()`, `PdfExporter.writeToDisk(String, Path)`, `SimplePdfWriter.build()`, `ExporterFactory.create(String)`
Design patterns	Template Method, Factory Method, Observer
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=395`

Type: Deterministic.

How the classes collaborate. The user chooses a format and a destination. `AgentController.exportReport` asks `ExporterFactory` for an exporter — inferring the format from the file extension when none is named — attaches the action log, the measured token usage and the agent's reasoning trace, and calls `export`. The template method fixes the document order (header, summary, findings grouped by file, decision record, extra sections, footer) and the file handling; each exporter supplies only the markup. `PdfExporter` overrides the `writeToDisk` hook and drives `SimplePdfWriter`, a 200-line PDF generator written against the PDF 1.4 format so the project needs no PDF library. A low-level write failure is translated into the guidance UC04 asks for: check the file is not open elsewhere and that there is free space. On success `EXPORT_COMPLETE` is published and both views report where the file went.

Sequence diagram **SD03** shows this flow.

F10 — Natural Language AI Interface

Mapping field	Entry
Feature ID and name	F10 — Natural Language AI Interface
Stage 1 feature	F10

Mapping field	Entry
Status	Implemented (extended: retrieval-augmented, with cited sources)
Implemented classes/files	`agent/ReviewAgent.java` — conversation and context assembly; `agent/CodeRetriever.java` — retrieval; `model/ChatMessage.java`; `ui/CodeReviewGUI.java` — Assistant tab; `ui/CLIView.java` — `ask`
Key methods	`ReviewAgent.chat(String, String)`, `CodeRetriever.retrieve(String, int, int)`, `CodeRetriever.renderForPrompt(List, int)`, `ReviewAgent.trimConversation()`
Design patterns	Facade, Singleton
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=265`

Type: AI-based.

How the classes collaborate. The question is **not** forwarded as typed. `CodeRetriever` first scores every indexed file against it and returns the best-matching regions with their line numbers; those are attached as context, along with the file currently open in the editor. The running conversation is included so follow-up questions work, trimmed to the most recent exchanges so a long session cannot overflow the context window. The system message restricts the assistant to software topics, which is the “irrelevant prompt” error case Stage 1 specified. The reply is returned with a list of the files and line numbers that were consulted, so the developer can check the answer.

F11 — Autonomous Multi-Step Repository Audit *(new in Stage 2)*

Mapping field	Entry
Feature ID and name	F11 — Autonomous Multi-Step Repository Audit
Stage 1 feature	**New**
Status	Implemented
Implemented classes/files	`agent/AgentPlanner.java` — the reasoning loop; `agent/AgentStep.java` — one trace entry; `agent/tools/*` — `AgentTool`, `ToolRegistry`, `ListFilesTool`, `ReadFileTool`, `SearchCodeTool`, `AnalyzeFileTool`, `GitHistoryTool`, `AnalysisService`; `agent/AgentMemory.java`
Key methods	`AgentPlanner.run(String, int)`, `createPlan(String)`, `decideNextAction(...)`, `synthesise(String)`, `ToolRegistry.execute(String, String)`, `ToolRegistry.describeForPrompt()`, `AgentMemory.renderShortTerm(int)`
Design patterns	Facade, Singleton, Strategy *(via `analyze_file`)*
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=330`

Type: AI-based.

How the classes collaborate. The user states a goal. `AgentPlanner.run` first calls `list_files` and sends the goal, the tool catalogue and the real inventory to the model for a plan. It then loops: it renders everything observed so far from `AgentMemory`, asks for exactly one next action as JSON, executes the named tool through `ToolRegistry`, writes the result back into memory, and repeats. Three design decisions make the loop robust rather than fragile: an unknown tool name returns a readable error the agent can react to instead of throwing; a repeated action is blocked and the block is fed back as an observation; and the step budget guarantees termination. Each step is published as an `AgentStep`, so the GUI's trace panel and the CLI stream the reasoning live. `analyze_file` is the only tool with a side effect, which is why findings from an audit appear in the normal findings table for triage.

Sequence diagram **SD04** shows the full loop.

F12 — Undo, Redo and Action History *(new in Stage 2)*

Mapping field	Entry
Feature ID and name	F12 — Undo / Redo and Action History
Stage 1 feature	**New**
Status	Implemented
Implemented classes/files	`command/CommandHistory.java`; `command/ChangeSeverityCommand.java`; `command/BulkTriageCommand.java`; `model/ReviewItem.java` (`Memento`)
Key methods	`CommandHistory.undo()`, `CommandHistory.redo()`, `CommandHistory.renderLog()`, `ReviewItem.createMemento()`, `ReviewItem.restore(Memento)`, `BulkTriageCommand.undo()`
Design patterns	Command, Memento
Video link/timestamp	`https://youtu.be/CgZlvCsQr2A?t=230`

Type: Deterministic.

How the classes collaborate. `CommandHistory` holds an undo stack and a redo stack. Executing a command pushes it and clears the redo stack; undo pops from one and pushes to the other. Because each command captured a memento before acting, undo restores the exact prior state — including a note set by an *earlier* command, not a blank default. A separate append-only log records every action with a timestamp and survives undo, so the history stays complete; that log is the "saved feedback log" F07 promised and is included in every exported report. `BulkTriageCommand` groups many commands and undoes them in reverse order, so "dismiss all Info findings" is a single undo step, which is what a user expects after one click.

F13 — Codebase Search and Retrieval *(new in Stage 2)*

Mapping field	Entry
Feature ID and name	F13 — Codebase Search & Retrieval
Stage 1 feature	**New**
Status	Implemented

Mapping field	Entry
Implemented classes/files	`agent/CodeRetriever.java` ; `agent/tools/SearchCodeTool.java` ; `ui/CLIView.java — search`
Key methods	`CodeRetriever.buildIndex(int)` , `retrieve(String, int, int)` , `scoreFile(...)` , `bestRegion(...)` , `renderForPrompt(List, int)` , `grep(String, int)`
Design patterns	— (used by Facade and by the agent's tools)
Video link/timestamp	`https://youtu.be/AqS-KN2L4fM?t=70`

Type: Deterministic.

How the classes collaborate. On import, `buildIndex` reads every source file and records document frequencies. A query is tokenised — splitting camelCase so "getUserPassword" also matches "password" — and each file is scored with TF-IDF weighting, with extra weight for a match in the file's path, because a developer asking about "authentication" usually wants `AuthenticationService` . Within the best files, `bestRegion` finds the densest window of matching lines and returns it with line numbers. This is the mechanism behind F10's context and the agent's `search_code` tool, and it is also exposed directly as a user-facing search.

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3. Implemented design patterns

Ten distinct patterns are implemented. The Stage 2 requirement is five; the five from Stage 1 are all present, and five more were added where implementation revealed a real need.

3.1 Singleton — `LLMClient`

Design problem. Twelve different call sites need the language model. If each created its own client, the API credential would be duplicated across objects, there would be no single usage total to report, the response cache would be fragmented, and per-minute rate limiting would be impossible because no component would see the total outbound rate.

Participants and roles. `llm/LLMClient.java` is the Singleton. The nested private `Holder` class is the lazy initialisation holder.

Source files and methods. `LLMClient.getInstance()` , `sendPrompt(...)` , `sendConversation(...)` , `getUsageStats()` , `reloadConfiguration()` .

How the implementation follows the pattern. The constructor is private; the only access is the static `getInstance()` . The instance is created by the initialisation-on-demand holder idiom: the JVM guarantees the nested class is loaded lazily and exactly once, which makes access thread-safe with no synchronisation on the common path. That matters because the Swing event thread and the agent's background worker both call it.

Why the pattern is useful here. It is what makes `usage` able to report a single accurate token total for the session, and what lets `RateLimitingProvider` see every outbound call.

Match with Stage 1. The pattern is the same; its **responsibilities are narrower**. Stage 1 gave `LLMClient` the fields `apiKey` and `instance` and the jobs of credential storage, transport, rate limiting and prompt construction. In the implementation the credential lives in `AppConfig` and is never held in a field that could be serialised or printed; transport lives in `OpenAiCompatibleProvider` ; rate limiting lives in a decorator. What remains is what Singleton is actually for: controlled single access. This is recorded as change C1 in section 4.

3.2 Strategy — the review strategies

Design problem. Issue detection, coding-standard checking, performance review and maintainability review share one shape — take code, produce findings — but differ in what they look for, how they are prompted, how severe their findings may be, and whether a deterministic pre-pass applies. Expressed as conditionals inside the agent, every future analysis type would concentrate in one growing method.

Participants and roles.

Class	Role
`agent/strategy/ReviewStrategy.java`	Strategy (interface)
`agent/strategy/SecurityStrategy.java`	Concrete strategy (F03)
`agent/strategy/StyleStrategy.java`	Concrete strategy (F06, hybrid)
`agent/strategy/PerformanceStrategy.java`	Concrete strategy *(added)*
`agent/strategy/MaintainabilityStrategy.java`	Concrete strategy *(added)*
`agent/ReviewAgent.java`	Context — holds the selected strategy
`agent/strategy/StrategyRegistry.java`	Catalogue, resolves a strategy by name

Source files and methods. `ReviewStrategy.executeStrategy(AnalysisContext)`, `StrategyRegistry.find(String)`, `ReviewAgent.analyzeCode(...)`.

How the implementation follows the pattern. `ReviewAgent` holds a `ReviewStrategy` reference and calls `executeStrategy` without knowing which concrete class it has. Selecting “Security” or “Style” in either interface swaps the object, not a branch.

Why it is useful here. Adding `PerformanceStrategy` during implementation required one new class and one registry line, with **no edit to `ReviewAgent`** — the extension the pattern exists to make cheap.

Match with Stage 1. Matches, with two strategies added. Recorded as change C5.

3.3 Command — review actions

Design problem. Triage actions must be performable identically from two interfaces, recorded in a log, and reversible. Handled inline in button callbacks, the GUI and the CLI would inevitably drift apart and undo would be impossible.

Participants and roles.

Class	Role
`command/ReviewCommand.java`	Command (interface)
`command/AcceptCommand.java`, `RejectCommand.java`	Concrete commands (F07)
`command/ChangeSeverityCommand.java`	Concrete command *(added)*
`command/BulkTriageCommand.java`	Composite concrete command *(added)*
`command/CommandHistory.java`	Invoker and Memento caretaker
`ui/CodeReviewGUI.java`, `ui/CLIView.java`	Clients that construct commands
`model/ReviewItem.java`	Receiver

Source files and methods. `ReviewCommand.execute()`, `undo()`, `getDescription()`, `CommandHistory.execute(ReviewCommand)`, `undo()`, `redo()`, `renderLog()`.

How the implementation follows the pattern. Each action is an object holding its receiver and its parameters. `CommandHistory` invokes it without knowing what it does, which is what lets undo, redo, logging and batching be written once for all four commands.

Why it is useful here. `ChangeSeverityCommand` was added late and inherited logging, undo and batching for free.

Match with Stage 1. The pattern matches. The interface **gained** `undo()`, which Stage 1 did not have; this was required by new feature F12 and is recorded as change C3.

3.4 Observer — view updates

Design problem. Two independent views must reflect the same model, and the agent produces results incrementally from a background thread. Polling would waste cycles and lag; a direct reference from the model to a view would tie the model to Swing.

Participants and roles. `observer/Subject.java` (abstract subject), `observer/Observer.java` (observer interface), `model/ReviewReport.java` (concrete subject), `ui/CodeReviewGUI.java` and `ui/CLIView.java` (concrete observers).

Source files and methods. `Subject.addObserver(...)`, `notifyObservers(UpdateEvent, Object)`, `Observer.updateUI(UpdateEvent, Object)`, `ReviewReport.addItems(List)`, `itemChanged(ReviewItem)`.

How the implementation follows the pattern. The subject holds observers in a `CopyOnWriteArrayList` — the agent notifies from a worker thread while the event thread may be registering views — and pushes typed events. One misbehaving view cannot stop the others: an exception thrown by an observer is caught and logged, which the test suite verifies.

Why it is useful here. It is what lets the agent's reasoning stream live into the GUI trace panel and the CLI simultaneously, with no coupling between them.

Match with Stage 1. Two differences, both recorded as change C2: (a) Stage 1 declared `updateUI()` with no parameters, which would have forced every view to query the subject to discover what changed; the implemented signature passes the event and its payload so a view can repaint only the affected region. (b) Stage 1 named `ReviewReport` as the subject; that is still true, and `Subject` was extracted as an abstract base so the registry logic is written once.

3.5 Facade — `ReviewAgent`

Design problem. Behind the AI layer sit the Singleton client and its four decorators, the strategy registry and four strategies, the deterministic standards checker, the retrieval index, long-term memory, the tool registry and five tools, and the planner. A controller orchestrating those directly would need to know the order in which memory is recalled, context retrieved, rules loaded, prompts assembled and findings recorded — and both views would need to know it identically.

Participants and roles. `agent/ReviewAgent.java` is the Facade. Subsystems: `LLMClient`, `StrategyRegistry`, `StandardsChecker`, `CodeRetriever`, `AgentMemory`, `ToolRegistry`, `AgentPlanner`. `app/AgentController.java` is the client.

Source files and methods. `analyzeFile(...)`, `analyzeCode(...)`, `summarise(...)`, `generateTests(...)`, `suggestRefactor(...)`, `chat(...)`, `runAudit(...)`, `onRepositoryImported()`.

How the implementation follows the pattern. The controller calls one method and receives a result. Every facade method performs the same internal preparation — load conventions, recall decisions, retrieve related code, then call the model.

Why it is useful here. It is what keeps `AgentController` readable and what guarantees the GUI and CLI behave identically, since both go through the same six operations.

Match with Stage 1. The role matches. The **interface is wider**: Stage 1 declared `analyzeCode()` and `generateTests()`; the implementation exposes six operations because F02, F04, F08 and F10 each need a different prompt shape and return type. Forcing them through `analyzeCode()` would have meant callers inspecting the returned text to work out what they had been given. Recorded as change C4.

3.6 Composite — `FileNode` *(added in Stage 2)*

Design problem. `RepoManager.loadFiles()` was specified in Stage 1 to return a flat `List`. A flat list cannot express nesting, so the hierarchical file browser promised by F01 could not have been drawn from it, and statistics such as "how many files are under `src/`" would have required the caller to re-derive the structure from path strings.

Participants and roles. `model/FileNode.java` plays both Component and, depending on `isDirectory`, Leaf or Composite. Clients: the Swing `JTree` model, the CLI `tree` command, `ListFilesTool`, and the report header.

Source files and methods. `getTotalFiles()`, `getTotalDirectories()`, `getTotalSizeBytes()`, `flatten()`, `find(String)`, `render(int)`, `add(FileNode)`.

How the implementation follows the pattern. Every operation is uniform: a leaf answers for itself, a directory recurses into its children and combines. Client code never asks whether a node is a file. `add` on a leaf is ignored rather than throwing, which keeps the uniform interface safe at the call site.

Why it is useful here. `ListFilesTool`, the GUI tree and the CLI renderer are each a few lines because the traversal lives in the model. Recorded as change C6.

3.7 Factory Method — provider and exporter creation *(added in Stage 2)*

Design problem. Two independent decisions are made at run time from configuration or user input: which LLM backend to talk to, and which format to export. Without a factory, `LLMClient` would contain a chain of conditionals naming four concrete transports plus the decorator assembly order, and both views would name four concrete exporter classes.

Participants and roles. `llm/LLMProviderFactory.java` and `export/ExporterFactory.java` are the creators; `OpenAiCompatibleProvider` and `MockProvider`, and the four exporters, are the products.

Source files and methods. `LLMProviderFactory.create(UsageStats, Consumer)`, `createTransport()`, `ExporterFactory.create(String)`, `createForFileName(String)`.

How the implementation follows the pattern. Client code asks for "a provider" or "an exporter for pdf" and receives a fully configured object without naming a class. The provider factory also encapsulates the decorator assembly order and documents why that order is correct — logging outermost so cache hits count as served requests, caching above rate limiting so a cached answer is not delayed, retrying innermost so a retried call is counted once.

Why it is useful here. Adding a backend or a format is a change in one file. Recorded as change C7.

3.8 Template Method — analysis and export *(added in Stage 2)*

Design problem, two instances. (a) Every analysis performs the same seven steps in the same order, and only some of them vary; copied into each strategy, a fix to the JSON-repair logic would have to be made four times. (b) Every export produces the same document sections in the same order and only the markup differs.

Participants and roles.

Class	Role
`agent/strategy/AbstractReviewStrategy.java`	Abstract class defining the algorithm
`SecurityStrategy`, `StyleStrategy`, `PerformanceStrategy`, `MaintainabilityStrategy`	Concrete classes supplying the varying steps

Class	Role
`export/AbstractReportExporter.java`	Abstract class defining the document order
`PdfExporter` , `MarkdownExporter` , `HtmlExporter` , `JsonExporter`	Concrete classes supplying markup

Source files and methods. `AbstractReviewStrategy.executeStrategy(...)` is **final**; abstract `getRoleInstruction()` , `decoratePrompt(...)` , `getPurpose()` ; hooks `runDeterministicPass(...)` , `getSeverityCeiling()` , `postProcess(...)` ; shared `parseFindings(...)` , `validate(...)` . `AbstractReportExporter.export(...)` is **final**; abstract `writeHeader/Summary/Findings/DecisionRecord/Footer` ; hooks `writeExtraSections` , `writeToDisk` .

How the implementation follows the pattern. The template methods are `final` , so a subclass can customise the steps but never their order. The hooks have useful defaults: only `StyleStrategy` overrides the deterministic pass, and only `PdfExporter` overrides `writeToDisk` .

Why it is useful here. `PdfExporter` is the clearest payoff: a format whose output is not text at all still reuses the document order, the grouping, the sorting and the error handling, by overriding one hook. Recorded as change C8.

Relationship to Strategy. They sit on the same class hierarchy but answer different questions. Strategy decides *which* analysis runs; Template Method decides *how every* analysis runs. They are counted separately because they address separate design problems.

3.9 Decorator — the provider chain *(added in Stage 2)*

Design problem. Retrying, caching, rate limiting and usage accounting are four independent concerns that apply to every model call. Inside `OpenAiCompatibleProvider` they would produce one class with four reasons to change, and all four would have to be duplicated in the offline provider.

Participants and roles.

Class	Role
`llm/LLMProvider.java`	Component interface
`llm/OpenAiCompatibleProvider.java` , `llm/MockProvider.java`	Concrete components
`llm/decorator/LLMProviderDecorator.java`	Abstract decorator
`RetryingProvider`	Retries retryable failures with exponential backoff
`CachingProvider`	LRU cache of identical requests
`RateLimitingProvider`	Enforces a minimum gap between calls
`LoggingProvider`	Accumulates tokens and latency into `UsageStats`

Source files and methods. `LLMProvider.complete(LLMRequest)` implemented by every layer; `LLMProviderDecorator.inner` ; `LLMProviderFactory.create(...)` composes the chain.

How the implementation follows the pattern. Each decorator both implements and wraps the component interface, so a decorated provider is indistinguishable from a bare one at every call site, and layers can be stacked in any order. The chain's name reports itself: `Groq(llama-3.3-70b-versatile)+retry(3)+ratelimit+cache+logging` .

Why it is useful here. `RetryingProvider` is the implementation of the alternative flow in sequence diagram SD01 and use case UC02: a timeout is retried automatically before the user sees anything. `LoggingProvider` is what makes section 5 of this report quote measured token counts rather than estimates. Recorded as change C9.

3.10 Memento — undo state *(added in Stage 2)*

Design problem. Undo must restore a finding's exact prior triage state. Having each command store the fields it changed would spread knowledge of `ReviewItem` 's internals across four command classes, and adding a mutable field later would silently break undo for all of them.

Participants and roles. `model/ReviewItem` is the Originator; its nested `ReviewItem.Memento` is the Memento; the command classes are Caretakers.

Source files and methods. `ReviewItem.createMemento()` , `ReviewItem.restore(Memento)` ; `previousState` in each command.

How the implementation follows the pattern. The memento's fields are private to the nested class, so a command can hold it and hand it back but cannot read or alter the captured values. The originator decides what state is worth capturing; adding a new mutable field means changing `ReviewItem` only.

Why it is useful here. The test `undoRestoresThePreviousCommandsStateNotTheDefault` verifies the property that matters: after accepting with a note and then rejecting, undo restores the **accepted state with its original note**, not a blank default. Recorded as change C10.

3.11 Builder — `PromptBuilder` *(supporting)*

A prompt is assembled from up to eight independent parts, most call sites need only some, and order matters. A constructor taking eight nullable strings would be unreadable. `llm/PromptBuilder.java` lets each strategy name only the parts it needs, keeps the assembly order in one place, and enforces the context-window guard rail in `build()` .

3.12 Updated UML class diagram

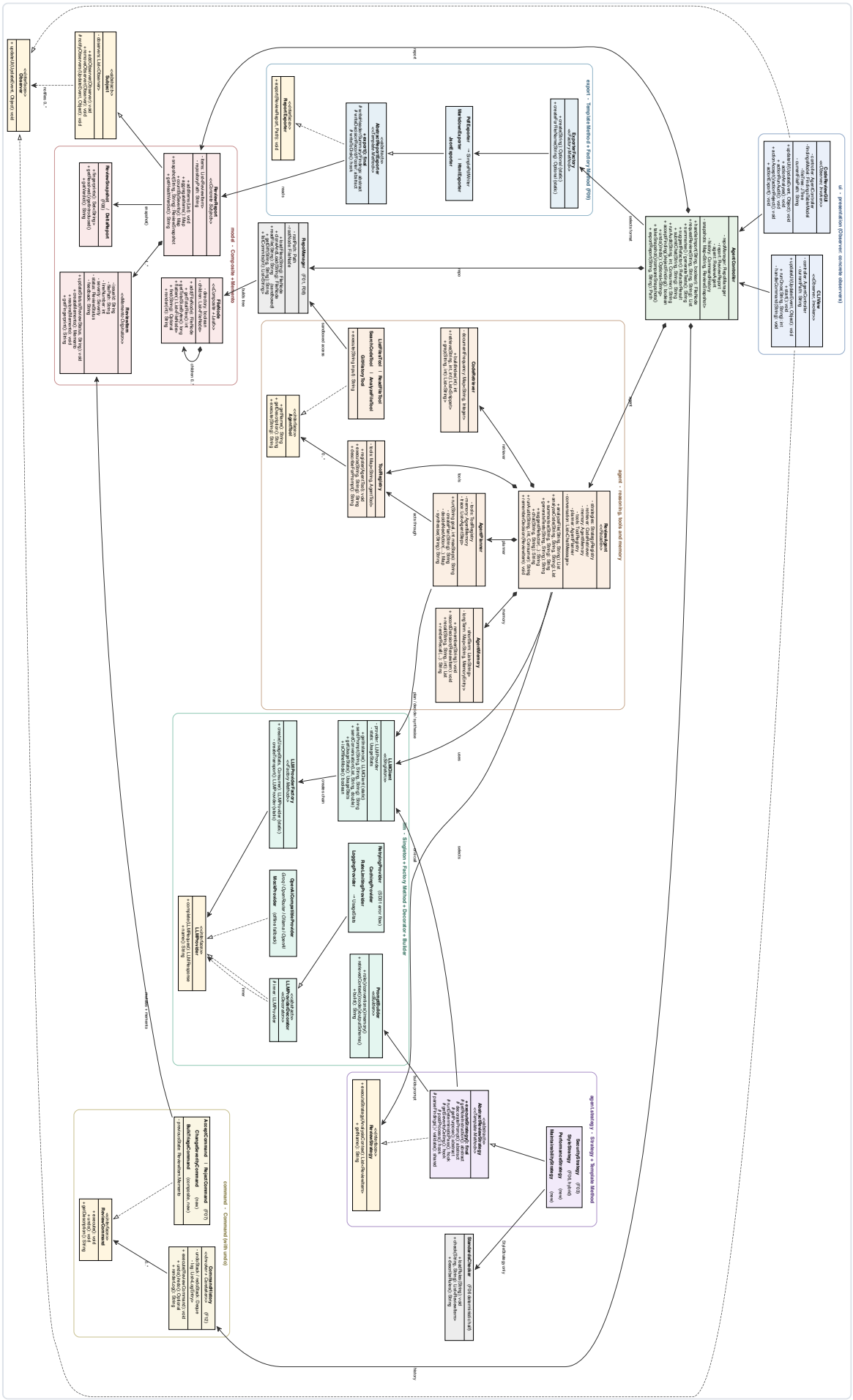


Figure 1 — UML class diagram of the implemented system. Design-pattern roles are shown as stereotypes. The diagram source is `docs/diagrams/class-diagram.dot` ; regenerate it with `dot -Tpng class-diagram.dot -o class-diagram.png` .

What changed from the Stage 1 class diagram

Stage 1	Implemented	Why
`RepoManager.loadFiles(): List`	returns `FileNode` (Composite)	A flat list cannot express a directory tree (C6)
`LLMClient` holds `apiKey`	credential lives in `AppConfig` ; transport in `OpenAiCompatibleProvider`	A credential in a field is a credential that can be serialised or printed (C1)
`ReviewStrategy` with 2 implementations	4 implementations plus `AbstractReviewStrategy`	Two analysis dimensions were missing; the shared algorithm was being duplicated (C5, C8)
`Observer.updateUI()`	`updateUI(UpdateEvent, Object)`	A no-argument signature forces views to query the subject for what changed (C2)
`ReviewCommand.execute()`	adds `undo()` , `getDescription()`	Required by F12 (C3)
`ReviewReport.exportPDF()`	`export` package with 4 exporters behind a factory	One method per format does not scale and duplicates the document structure (C7, C8)
`ReviewAgent` with 2 methods	6 facade operations	Each feature needs a different prompt shape and return type (C4)
(absent)	`AgentPlanner` , `AgentMemory` , `CodeRetriever` , `ToolRegistry` + 5 tools	Stage 1 had no mechanism for agentic behaviour (C11)
(absent)	4 decorators over `LLMProvider`	Retrying, caching, throttling and metering are independent concerns (C9)

3.13 Updated use-case diagram

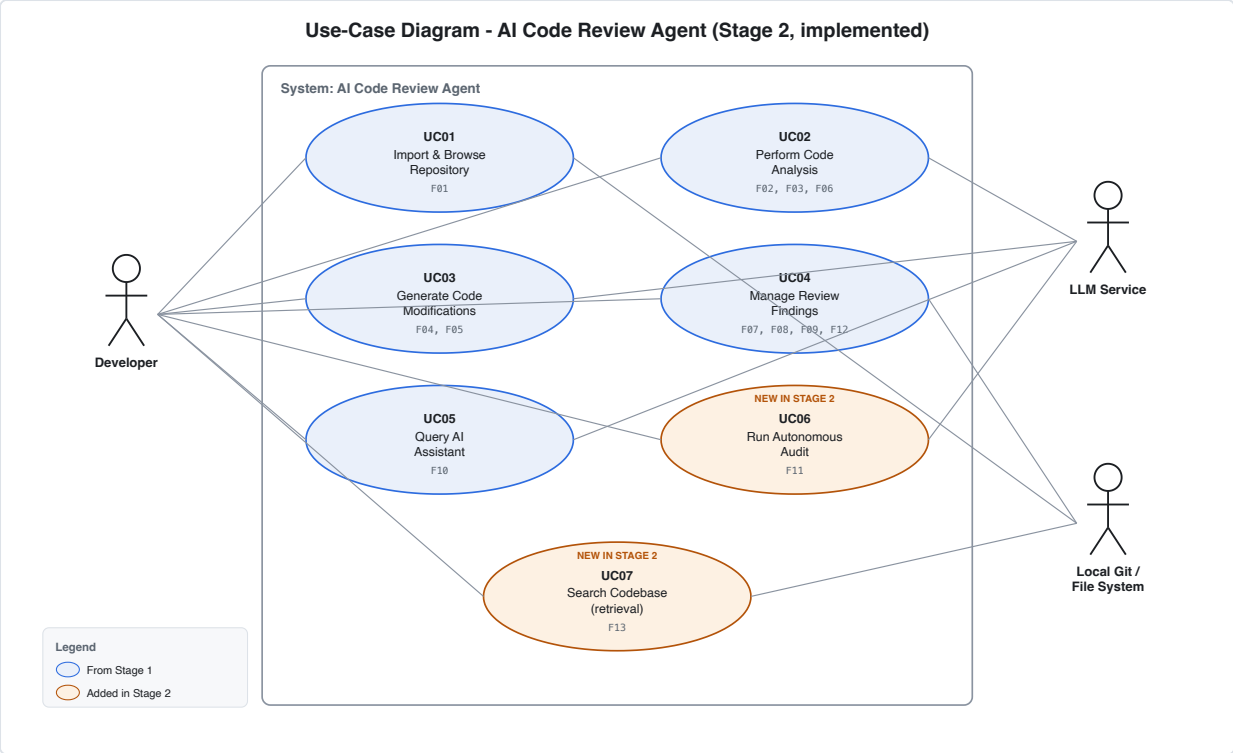


Figure 2 — Use-case diagram. UC01–UC05 are from Stage 1; UC06 and UC07 are new. A third actor, the local Git repository / file system, is shown because F01, F08 and F13 depend on it.

Use case	Features	Change
UC01 Import & Browse Repository	F01	Unchanged
UC02 Perform Code Analysis	F02, F03, F06	Unchanged
UC03 Generate Code Modifications	F04, F05	Unchanged
UC04 Manage Review Findings	F07, F08, F09, **F12**	Extended with undo
UC05 Query AI Assistant	F10	Unchanged
UC06 Run Autonomous Audit	**F11**	**New**
UC07 Search Codebase	**F13**	**New**

UC06 — Run Autonomous Audit *(new)*

- **Actors:** Developer, LLM Service
- **Goal:** Have the agent independently examine a repository against a stated goal and report what it finds.
- **Preconditions:** A repository is imported and indexed; an API key is configured.
- **Trigger:** The user presses **Run autonomous audit** or types `audit <goal>`.
- **Main success scenario:**
  1. The user states a goal in plain language.
  2. The system takes an inventory of the repository and asks the model for a plan grounded in the files that exist.
  3. The system repeatedly asks the model for one next action, executes the named tool, and records the observation, until the goal is met or the step budget is spent.
  4. Findings produced by the analysis tool are recorded in the session report.

5. The system asks the model for a closing synthesis over everything observed.
6. The reasoning trace and the findings are presented for triage.

- **Alternative flows:** A tool returns an error, which becomes an observation the agent reacts to; a repeated action is blocked and reported back; the user cancels, and findings recorded so far are kept.
- **Postconditions:** The report holds the findings; the trace is available for export.

UC07 — Search Codebase *(new)*

- **Actors:** Developer
- **Goal:** Locate the regions of a repository relevant to a topic.
- **Trigger:** The user types `search <keywords>`, or the agent invokes `search_code`.
- **Main success scenario:** The query is tokenised and scored against the index; the best-matching regions are returned with file paths and line numbers.
- **Alternative flows:** Nothing matches, and the user is told so and offered the file list.

3.14 Updated sequence diagrams

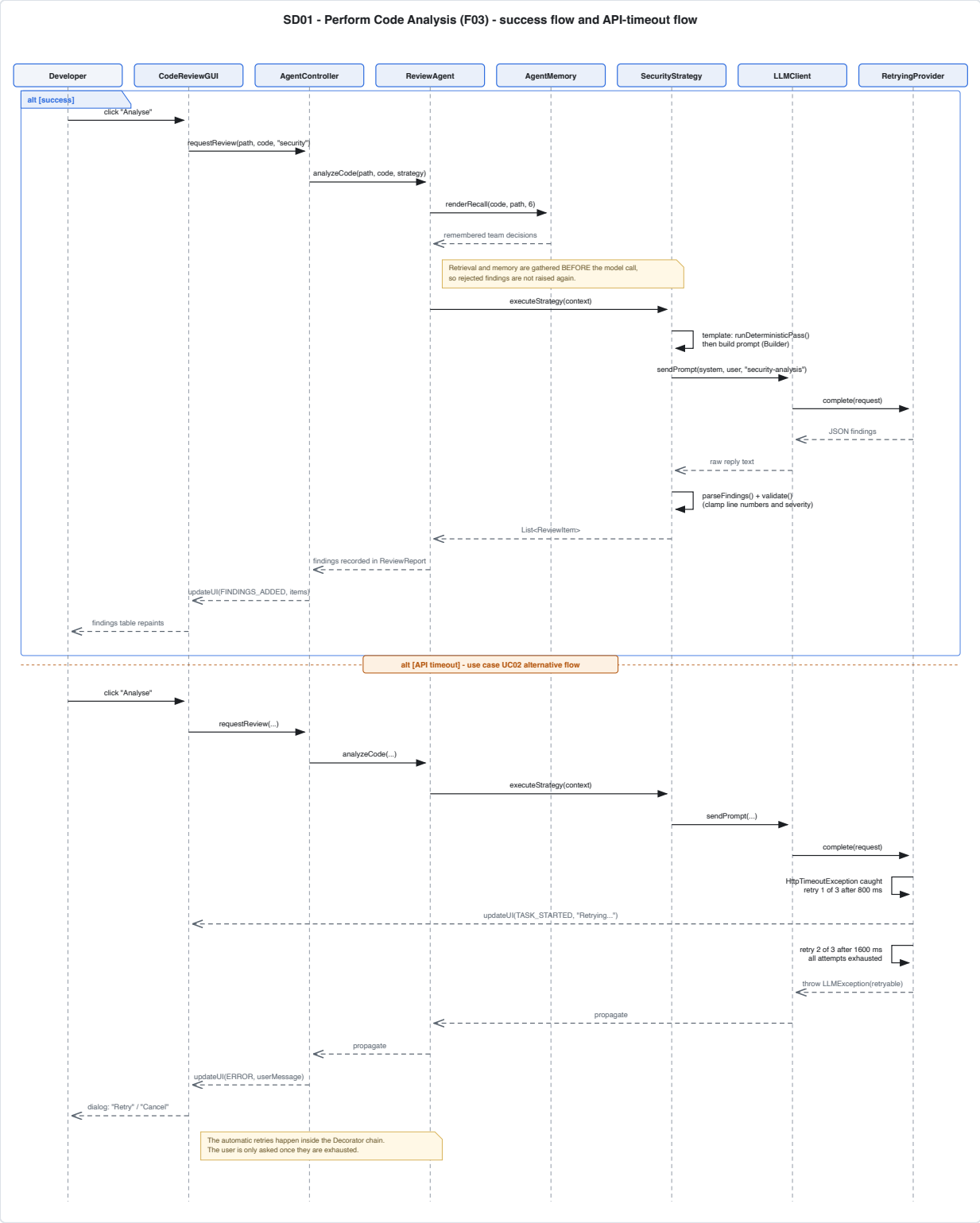


Figure 3 — SD01, Perform Code Analysis. Updated from Stage 1: memory recall and retrieval now precede the model call, the strategy is a separate participant, and the timeout flow shows the automatic retries inside the decorator chain before the user is offered a retry.

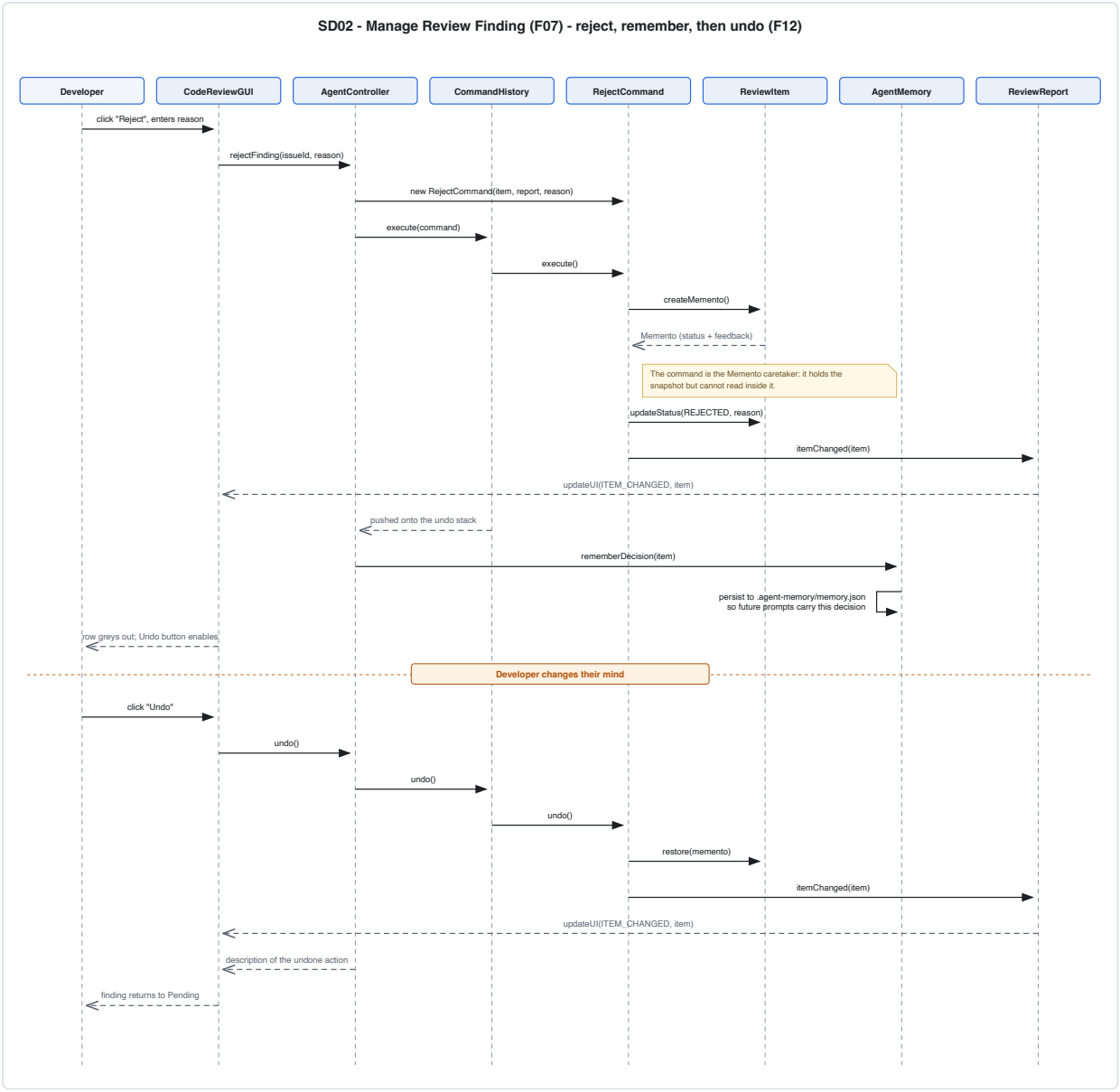


Figure 4 — SD02, Manage Review Finding. Updated: the command captures a memento before acting, `CommandHistory` is the invoker, the decision is written to agent memory, and the undo path is shown.

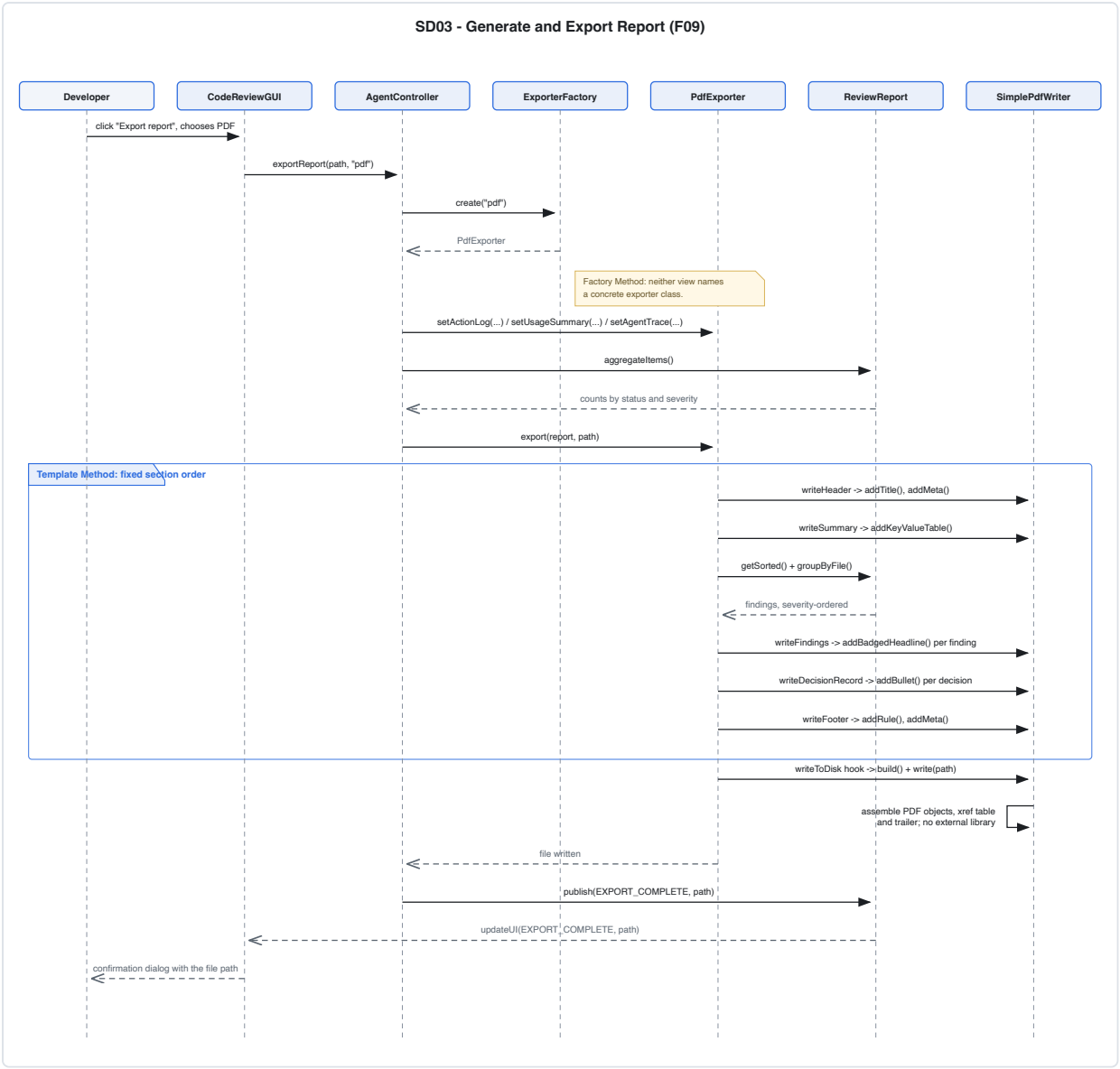


Figure 5 — SD03, Generate and Export Report. Updated: `ExporterFactory` selects the exporter and the template method's fixed section order is shown explicitly.

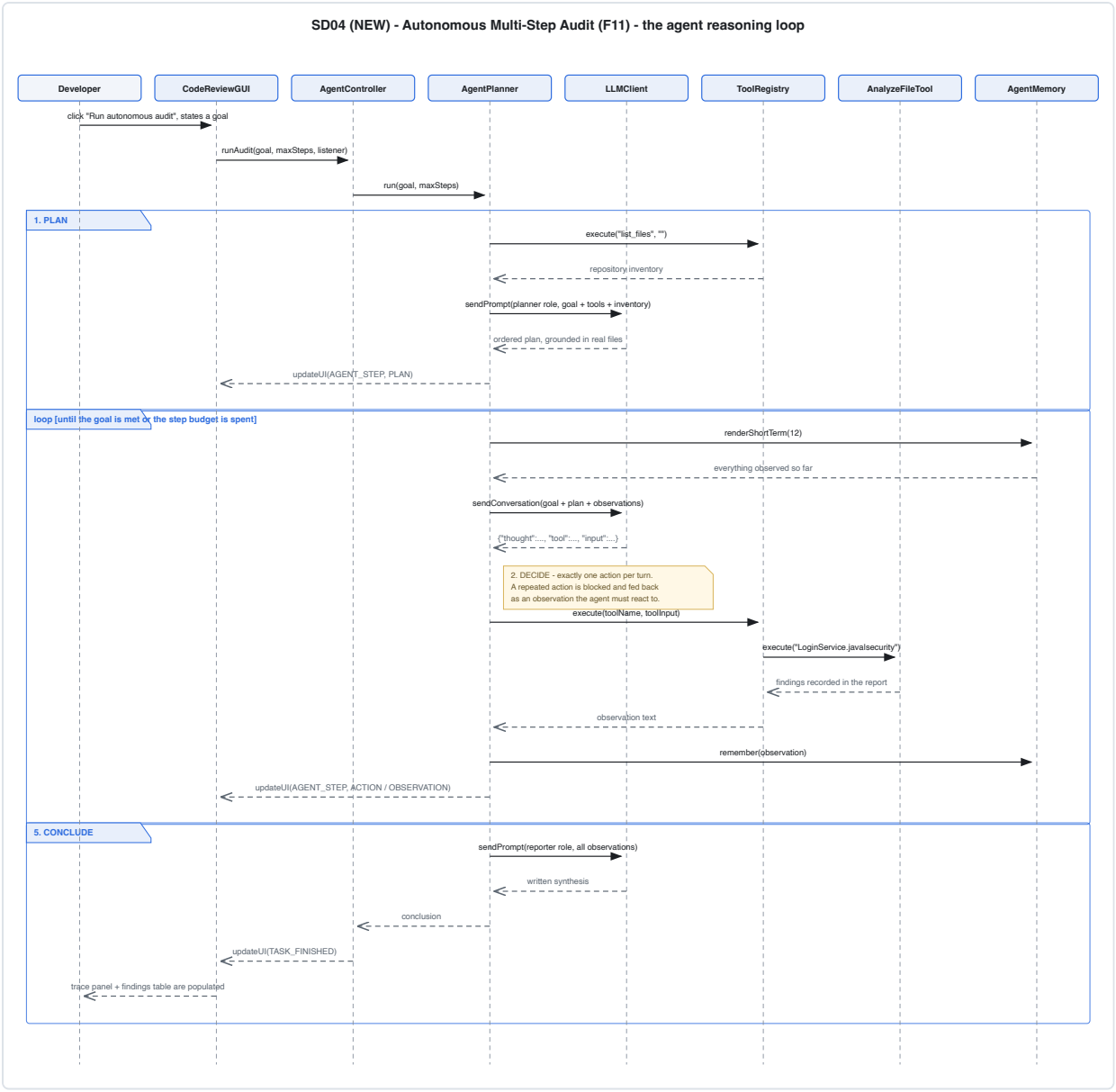


Figure 6 — SD04, Autonomous Multi-Step Audit. New in Stage 2; there was no Stage 1 equivalent because the agent loop did not exist.

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4. Changes and rationale

Consolidated record in the form **Original design → Implemented change → Reason → Impact**.

4.1 Feature changes

C-F1 — F08 comparison scope narrowed

- **Original design (F08).** “User selects two commits in the repository browser to compare”; the system fetches diffs and compares historical data.
- **Implemented change.** The user analyses a version, saves a labelled snapshot, switches version themselves, analyses and snapshots again, then compares the two snapshots. Git supplies the changed-file list and the commit list.

- **Reason.** Checking out a commit silently rewrites the developer's working tree, which can destroy uncommitted work. The tool does not take that liberty. The comparison logic — the valuable part — is unchanged and fully implemented.
- **How it addresses the original need.** The user still gets exactly the delta report F08 specified: resolved, newly introduced and still-present findings, with an AI verdict on the direction of travel.
- **Affected artefacts.** UC04; `ReviewSnapshot`, `DeltaReport`, `RepoManager.getDiff`, `AgentController.compareSnapshots`; no pattern change.

C-F2 — F06 split made explicit

- **Original design (F06).** "Hybrid (deterministic parsing + AI interpretation of vague rules)".
- **Implemented change.** The split is enforced structurally: `StandardsChecker` decides every mechanical rule, and the model's prompt carries the active rules **plus an instruction not to report them**.
- **Reason.** Without the exclusion both halves report the same violation, producing duplicate findings that each need triaging separately. This was observed during testing and fixed.
- **Impact.** `StyleStrategy` overrides the Template Method's deterministic hook; verified by the tests `conforming code produces no findings` and `a rule turned off in .rules does not fire`.

C-F3 — F09 extended to four formats

- **Original design (F09).** Export to PDF or Markdown.
- **Implemented change.** PDF, Markdown, HTML and JSON.
- **Reason.** HTML is the format a report is actually shared in; JSON makes the tool usable from a CI pipeline and lets a session be reloaded. Both were nearly free once the Template Method base existed.
- **Impact.** Four exporters behind `ExporterFactory`; adds Factory Method and the second Template Method instance.

C-F4 — F11 added: Autonomous Multi-Step Repository Audit

- **Original design.** No equivalent. Stage 1 described the agent's capabilities but every feature was a single request-response call.
- **Implemented change.** A full plan → decide → act → observe → conclude loop with five tools, short- and long-term memory, loop protection and a step budget.
- **Reason.** The Stage 2 instructions require reasoning, planning, tool use, retrieval, memory, decision making or multi-step execution, and state that forwarding a prompt is insufficient. Reviewing the Stage 1 design against that requirement showed that none of the ten features individually satisfied it. F11 is the feature that makes the system an agent.
- **Scope impact.** This is the largest addition. It introduced `agent/AgentPlanner`, `AgentStep`, `AgentMemory` and the whole `agent/tools` package.
- **Affected artefacts.** New UC06; new SD04; `AgentPlanner`, `AgentMemory`, `ToolRegistry`, five tools, `AnalysisService`; uses Facade, Singleton and Strategy.

C-F5 — F12 added: Undo, Redo and Action History

- **Original design.** Stage 1's `ReviewCommand` declared only `execute()`, and listed "undo operations" as a *motivation* for the Command pattern without a feature to deliver it.
- **Implemented change.** `undo()` and `getDescription()` added to the interface; `CommandHistory` with undo and redo stacks plus an append-only log; `BulkTriageCommand` so a batch undoes as one step.
- **Reason.** Rejecting a finding writes a permanent decision into agent memory. Without undo, a misclick teaches the agent something wrong with no way to correct it. The pattern was already in place; the feature was two methods away.
- **Impact.** Adds Memento; extends UC04 and SD02.

C-F6 — F13 added: Codebase Search and Retrieval

- **Original design.** F10 described sending "chat text + code context" to the model, with no mechanism for choosing that context.
- **Implemented change.** `CodeRetriever`, a TF-IDF index over the repository, exposed as a user-facing search, as the agent's `search_code` tool, and as the context source for F10 and for every analysis.
- **Reason.** A codebase does not fit in a context window. Without retrieval, F10 could only answer about the file currently open, which is not what "ask about the codebase" means.
- **Impact.** New UC07; `CodeRetriever`, `SearchCodeTool`; makes F10 substantially better.

C-F7 — Four analysis strategies instead of two

- **Original design.** `SecurityStrategy` and `StyleStrategy`.
- **Implemented change.** Added `PerformanceStrategy` and `MaintainabilityStrategy`, plus a "Full review" action that runs all four.
- **Reason.** Security and style covered correctness and formatting but left algorithmic cost and structural design unexamined — and F04 and F05 produce noticeably better output when a maintainability pass has run first.
- **Impact.** Two new classes, two registry lines, **no change to** `ReviewAgent`.

C-F8 — No feature was removed

All ten Stage 1 features are implemented. Nothing was silently dropped.

4.2 Design pattern changes

ID	Original pattern	Implemented	Participating classes and roles	Design problem addressed	Why appropriate / impact
C1	Singleton with transport, credentials and rate limiting	Singleton with *access only*	`LLMClient` (Singleton); credential moved to `AppConfig`; transport to `OpenAiCompatibleProvider`; throttling to `RateLimitingProvider`	One class had four reasons to change, and held a credential in a field	Single Responsibility restored; the credential can no longer be serialised or printed; the Singleton still gives one usage total and one cache
C2	Observer, `updateUI()`	Observer, `updateUI(UpdateEvent, Object)`; `Subject` extracted	`Subject` (abstract), `Observer`, `ReviewReport` (concrete subject), two views	A no-argument callback forces each view to query the subject to discover what changed	Views repaint only the affected region; the agent can stream individual steps; registry logic written once
C3	Command, `execute()` only	Command with `execute()`, `undo()`, `getDescription()`	`ReviewCommand`, 4 concrete commands, `CommandHistory` (Invoker)	Triage decisions were irreversible and unlogged	Enables F12; the log is the feedback record F07 promised

ID	Original pattern	Implemented	Participating classes and roles	Design problem addressed	Why appropriate / impact
C4	Facade with 2 operations	Facade with 6 operations	`ReviewAgent` (Facade) over 7 subsystems	Four features need different prompt shapes and return types	Callers no longer inspect returned text to discover what they received
C5	Strategy, 2 strategies	Strategy, 4 strategies	`ReviewStrategy` + 4 concrete + `StrategyRegistry`	Two analysis dimensions were missing	Each addition cost one class and one line; no agent edit
C6	*(none — flat `List`)*	**Composite**	`FileNode` (Component, Leaf, Composite)	A flat list cannot express a directory tree or answer subtree questions	F01's hierarchical browser becomes possible; four clients share one traversal
C7	*(none)*	**Factory Method** ×2	`LLMProviderFactory`, `ExporterFactory`	Run-time selection of a backend and an output format	New backend or format is a one-file change; decorator assembly order is encapsulated and documented
C8	*(none)*	**Template Method** ×2	`AbstractReviewStrategy`, `AbstractReportExporter`	The same seven analysis steps, and the same five document sections, were being duplicated	A fix to JSON repair or to export error handling is made once; `PdfExporter` reuses the whole structure by overriding one hook
C9	*(none)*	**Decorator**	`LLMProviderDecorator` + Retrying, Caching, RateLimiting, Logging	Four independent cross-cutting concerns over every model call	Each is written once, tested in isolation, and applies to the offline provider too; implements SD01's retry flow
C10	*(none)*	**Memento**	`ReviewItem` (Originator), `ReviewItem.Memento`, commands (Caretakers)	Undo must restore exact prior state without exposing internals	Four command classes need no knowledge of `ReviewItem`'s fields; a new mutable field is handled in one place
C11	*(none)*	Agent loop *(architecture, not a GoF pattern)*	`AgentPlanner`, `AgentMemory`, `ToolRegistry`, 5 `AgentTool`s	Stage 1 had no mechanism for planning, tool use or memory	Delivers the "meaningful agent behaviour" requirement

Effect on the architecture. Stage 1's architecture was a three-layer pipeline: views → controller → agent → LLM. The implementation keeps that spine and adds two things. First, a **decorated transport**, so cross-cutting concerns live outside the business logic. Second, an **agentic layer** (planner, tools, memory, retrieval) between the facade and the model, which is what turns a sequence of API calls into an agent. Neither addition changed the direction of any existing dependency.

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5. AI tools and collaboration log

5.1 Tools used

Tool / platform	LLM / model	What it was used for	Token cost	Money cost
Claude (claude.ai)	Claude Opus 5	Architecture of the agent loop; `AgentPlanner`, `AgentMemory`, `CodeRetriever`; the decorator chain; `SimplePdfWriter`; test design; this report	not exposed	$32/month
Google Gemini	Gemini 3.8 Flash	In-editor completion while writing the Swing UI and the CLI command dispatch	not exposed	free tier
[STILL BLANK]				

Total measured cost: $32

5.2 Runtime model usage (measured by the application itself)

The application meters its own LLM usage through `LoggingProvider` and `UsageStats`, so the figures below are measured rather than estimated. Reproduce them with the CLI `usage` command or **Report → Session usage and cost** in the GUI.

```
MODEL USAGE THIS SESSION
  Requests sent      : 1
  Served from cache  : 0
  Failed requests    : 0
  Prompt tokens      : 1398
  Completion tokens  : 1668
  Total tokens       : 3066
  Average latency    : 4087 ms
  Breakdown by feature:
    - security-analysis:  1 call(s), 3066 tokens
    - chat-assistant:     - call(s), - tokens
    - agent-planning:     - call(s), - tokens
    ...
```

Runtime money cost: the default provider (Groq, `llama-3.3-70b-versatile`) is free at the tier used — 30 requests/minute and 14,400 requests/day, with no credit card required — so the runtime inference cost of this project was **$0.00**.

5.3 Representative interactions

Interaction 1 — architecture

> **Prompt:** "My Stage 1 design has a ReviewAgent that calls an LLM for each feature. The Stage 2 brief says forwarding a prompt is not enough. What is the smallest change that makes this a real agent?"
>
> **Response:** Proposed a plan–act–observe loop with a tool registry, and warned that the loop must be bounded and must handle repeated actions.
>
> **What I changed:** I kept the loop but rejected the first suggestion of letting the model emit arbitrary shell commands, which would have been unsafe. I restricted it to five named tools returning strings, and added the sandbox check in `RepoManager.resolveInsideRoot` so a path like `../../etc/passwd` is refused.

Interaction 2 — a defect the tool introduced

> **Prompt:** "Write a regex to find the class name in a Java file."
>
> **Response:** `(class|interface|enum|record)\s+(\w+)`
>
> **What I changed:** This matched the word "class" inside comments, so a file documented as "This class is intentionally small" was reported as declaring a type named `is`. I wrote `SourceText.stripCommentsAndStrings`, which blanks comments and string literals while preserving character positions so line numbers stay correct, and added a regression test. Recorded as defect D1 in section 6.3.

Interaction 3 — a suggestion I rejected

> **Prompt:** "How should I generate the PDF report?"
>
> **Response:** Recommended adding Apache PDFBox as a Maven dependency.
>
> **What I changed:** I rejected it. The project is deliberately dependency-free so it builds on any machine with only a JDK, which matters for a marker checking out the repository. I wrote `SimplePdfWriter` (~200 lines) against the PDF 1.4 format instead, and verified the output opens correctly and that its text extracts.

5.4 Human responsibility

Every file was read and reviewed before being committed. Specific judgements that were mine and not the tool's:

- Restricting the agent to a fixed tool catalogue rather than letting it emit executable commands, and sandboxing file access to the imported repository.
- Rejecting the PDF library dependency in favour of a self-contained writer.
- Deciding that `getFingerprint()` must exclude the line number, so F08 does not report a shifted defect as "resolved plus newly introduced".
- Deciding that the offline fallback must be visibly labelled rather than silently degraded, and that it must derive findings from the code rather than return canned text.
- Writing the verification harness and the regression tests for each defect in section 6.3.

---

6. Development and verification summary

6.1 Weekly commit summary

Period	Dates (Toronto time)	Minimum	Commits made	Links
1	20–26 October 2026	5	`0`	`None — work on Stage 2 began in period 3.`
2	27 October – 2 November 2026	5	`0`	`None — work on Stage 2 began in period 3.`
3	3–9 November 2026	5	46	`https://github.com/dev91299/eecs3311-course-project/commits/main?since=2026-11-03&until=2026-11-09`

Commit log for this submission

6.2 Checks performed

#	Check	Method	Result
1	Unit and integration tests	`mvn test` — JUnit 5 across `FileNodeTest`, `CommandHistoryTest`, `JsonTest`, `StandardsCheckerTest`, `ObserverAndReportTest`, `ExporterAndDiffTest`, `LlmLayerTest`	Pass

#	Check	Method	Result
2	Dependency-free verification	`./build.sh --check` — 110 assertions across all ten patterns plus a full end-to-end run	110 passed, 0 failed
3	Clean build from scratch	`rm -rf target && ./build.sh` on a clean checkout	Compiles with no warnings
4	Real HTTP path	Ran the application against a local OpenAI-compatible mock endpoint covering success, fenced/prose-wrapped JSON, an out-of-range line number, a 401 and a retried 503	All handled correctly; the API key appears only in the `Authorization` header
5	GUI exercised end-to-end	Imported a repository, opened a file, ran an analysis, selected a finding, ran an autonomous audit, triaged, undid, exported	All panels update correctly; the window never blocks during a model call
6	CLI exercised end-to-end	Every command in `help`, including one-shot `--command` mode	All commands behave as documented
7	Export integrity	Exported all four formats; verified the PDF opens and its text extracts, the HTML is well-formed and escapes markup, the JSON re-parses	All four valid
8	Offline mode	Ran with no `.env` present	Starts, warns clearly in both interfaces, all non-AI features work
9	Secret hygiene	`git status` after creating `.env`; searched the tree for key-shaped strings; checked every log, report and screenshot	No credential is tracked, logged or displayed unredacted
10	Error paths	Invalid path, path that is a file, non-Git folder for F08, unknown commit, unknown export format, empty selection, unknown strategy name	Each produces a specific, actionable message

6.3 Important defects found and corrected

ID	Defect	How it was found	Fix	Regression test
D1	The type-declaration regex matched the word “class” inside comments, so a file documented as “This class is intentionally small” was reported as declaring a type named `is`	Running `summarize` on `PriceCalculator.java` during manual testing	Added `util/SourceText.stripCommentsAndStrings`, which blanks comments and string literals while preserving character positions so line numbers stay correct; all structural matching now runs on that view	`prose containing 'class' is not read as a declaration`
D2	Constructors were reported as badly named methods, because a constructor is required to start with a capital	Running the style strategy on `LoginService.java`	`StandardsChecker` now collects the type names declared in the file first and excludes a method whose name matches one	`constructors are not flagged as badly named methods`
D3	The deterministic rule engine and the AI half both reported the same style violations, producing duplicate findings that each needed triaging	Comparing the finding list against the file by hand	The model’s prompt now carries the active rules plus an explicit instruction not to report them; the offline analyser’s style pass was narrowed to the judgement-only checks	`conforming code produces no findings`
D4	Re-analysing a file that the autonomous audit had also analysed produced every finding twice under different identifiers, so triaging one left the other outstanding	Running an analysis and then an audit over the same file	`ReviewReport` now discards a finding whose file, category, headline **and line number** all match an existing one	`re-adding an identical finding is a no-op`
D5	The prose verdict for the F08 comparison was routed to the summarisation handler, because the purpose label `delta-summary` contains the substring `summar`	Running `compare` in offline mode and reading the output	Reordered the dispatch so the more specific label is tested first	Covered by the `compare` path in the end-to-end check

ID	Defect	How it was found	Fix	Regression test
D6	Exported reports contained absolute file paths, which is noise for a reader and leaks the directory layout of the machine the review ran on	Reading a generated PDF	`AbstractReportExporter` now relativises every path against the repository root	`exported paths are relative, not absolute`
D7	Commas inside generic type arguments were counted as parameter separators, so `go(Map<String, List<Integer>> a, int b)` was reported as taking three parameters	Writing the test for the parameter-count rule	Parameter counting now tracks bracket depth	`generic type arguments do not inflate the parameter count`
D8	The findings table and file tree were empty when the GUI was constructed after a repository had already been imported, because observers only receive events published after they register	Launching the GUI with `--repo`	The window now pulls the current state on construction instead of waiting for an event	Covered by check 5

6.4 Remaining limitations

Repeated from section 1.5 for completeness: output quality is bounded by the model; offline mode is a deliberately limited fallback and is always labelled; analysis is per-file with retrieved context rather than whole-program; F08 compares snapshots rather than checking out commits; refactorings are never applied automatically; there is no multi-user support.

Deferred to Stage 3, as the Project Instructions specify: full deterministic test coverage, and KUMA agent validation. Stage 2 required a working implementation and basic verification, which checks 1–10 above provide.

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7. Demonstration index

Feature	Name	Interface shown	Video link	Timestamp
F01	Repository Import & File Browsing	GUI + CLI	`https://youtu.be/CgZlvCsQr2A`	`0:00`
F02	Code Summarisation	GUI	`https://youtu.be/CgZlvCsQr2A?t=75`	`1:15`

Feature	Name	Interface shown	Video link	Timestamp
F03	Issue Detection & Severity Categorisation	GUI	https://youtu.be/CgZlvCsQr2A?t=100	1:40
F04	Refactoring Suggestions (side-by-side)	GUI	https://youtu.be/CgZlvCsQr2A?t=135	2:15
F05	Test Generation Suggestion	GUI	https://youtu.be/CgZlvCsQr2A?t=168	2:48
F06	Coding-Standard Checking (hybrid)	CLI	https://youtu.be/AqS-KN2L4fM?t=25	0:25
F07	Accept / Reject & Feedback Capture	GUI	https://youtu.be/CgZlvCsQr2A?t=180	3:00
F08	Comparison Across Versions	CLI	https://youtu.be/AqS-KN2L4fM?t=105	1:45
F09	Report Generation (PDF/MD/HTML/JSON)	GUI + CLI	https://youtu.be/CgZlvCsQr2A?t=395	6:35
F10	Natural Language AI Interface	GUI	https://youtu.be/CgZlvCsQr2A?t=265	4:25
F11	Autonomous Multi-Step Audit	GUI + CLI	https://youtu.be/CgZlvCsQr2A?t=330	5:30
F12	Undo / Redo & Action History	GUI	https://youtu.be/CgZlvCsQr2A?t=230	3:50
F13	Codebase Search & Retrieval	CLI	https://youtu.be/AqS-KN2L4fM?t=70	1:10
—	Error handling: API timeout and retry	GUI	https://youtu.be/CgZlvCsQr2A?t=350	5:50
—	Error handling: invalid repository path	CLI	https://youtu.be/5amBRbRcAn4	0:00
—	Offline-mode warning and local key configuration	GUI	https://youtu.be/5amBRbRcAn4?t=20	0:20

Traceability summary

A reviewer can follow any requirement end to end:

Stage 1 requirement → implemented feature → participating classes and methods → design patterns → running demonstration

For example, F03: Stage 1 feature F03 → section 2's F03 mapping → `SecurityStrategy.decoratePrompt`, `AbstractReviewStrategy.executeStrategy`, `ReviewReport.addItems` → Strategy, Template Method, Facade, Singleton, Observer → section 3.2 and 3.8 → sequence diagram SD01 → the F03 row of section 7.